PAPER_1554_M_W_80_379

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PAPER_1554 — W Boson Mass = $m_W = A_5 + 2\cdot SO_5 + F_{TRZ}\cdot D - F_{TRZ}^2\cdot D_{BSFG} + F_{TRZ}^2\cdot D - F_{TRZ}^2\cdot SSQ^2 \approx 80.38$ GeV

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Particle Physics
Date: June 18, 2026
Status: CLOSED — 0.003% closure from PAPER_1209HH_S653

Observation

PAPER_1209HH_S653 (Particle Physics Unified Proof Set) lists **W Boson Mass** with the integer-primitive expression:

$$\text{W Boson Mass} = m_W = A_5 + 2\cdot SO_5 + F_{TRZ}\cdot D - F_{TRZ}^2\cdot D_{BSFG} + F_{TRZ}^2\cdot D - F_{TRZ}^2\cdot SSQ^2 \approx 80.38\ \text{GeV}$$

Status: **0.003%**.

UQFF Closed Identity

$$m_W = A_5 + 2\cdot SO_5 + F_{TRZ}\cdot D - F_{TRZ}^2\cdot D_{BSFG} + F_{TRZ}^2\cdot D - F_{TRZ}^2\cdot SSQ^2 \approx 80.38\ \text{GeV}\ 0.003\%$$

Physical Interpretation

Tier-best closure in PAPER_1209HH at 0.003% residual — the most precisely matched electroweak mass.

Dominant: $A_5 + 2\cdot SO_5 = 60 + 20 = 80$ GeV — sits right at the W mass.
Corrections: tiny F_{TRZ}-suppressed terms reaching 80.38 GeV vs CODATA 80.379 GeV.

NOT REPLACEMENT

CODATA/PDG treats W Boson Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S653
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_w_80_379"})`
